SUPPORTING ADULT GUIDANCE

When two lines meet

Straight Lines and Connected Representations. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

The learner finds and interprets the **intersection** of two lines. Two lines with different gradients meet at exactly one point; in a cost story that point is the **break-even**, where the two options cost the same. The emphasis is on reading the crossing (its x and y) and explaining what it means — which option wins on each side.

Driving Question: *What does the point where two lines cross tell us?*

The mathematics, in plain terms

Two straight lines with **different gradients** cross at exactly one point, called the **intersection**. At that point the two lines share the same x and the same y, so in a cost story **both options cost exactly the same** — this is the **break-even point**. Reading it has two parts: the x tells you the number of units (visits, prints, games), and the y tells you the cost there. For example, a printer costing £30 plus £1 per ink pack ($y = x + 30$) versus a subscription of £4 per pack ($y = 4x$) breaks even where $x + 30 = 4x$, that is at $x = 10$ **packs**, both costing £40. The most useful idea is what the crossing means for a decision: **before** the break-even the option with the smaller starting cost is cheaper; **after** it, the option with the smaller rate is cheaper. So for a few packs the £0-upfront subscription wins, and for many packs buying the printer wins — the break-even is the tipping point where the best choice swaps over. This is exactly how businesses decide when a product starts to make a profit (where an income line overtakes a cost line). Learners can find the crossing by reading a graph, or by setting the two costs equal and solving. The next lesson uses lines to convert one quantity into another.

The crossing is a decision

The heart of this lesson is that a single point carries a real meaning: at the break-even the two options are equal, and it is the moment the cheaper choice changes hands. It is worth being explicit that the crossing has two coordinates doing different jobs — the x is 'how many', the y is 'the cost' — because learners often report only one. Protect the reasoning about the two sides: below the break-even the low-start option wins, above it the low-rate option wins. Grounding this in a genuine decision (buy or subscribe, membership or pay-per-visit) keeps it from being an abstract dot on a graph and gives the learner something they can actually use. Reading the crossing off a graph is enough at this stage; setting the costs equal and solving is a natural next step for those ready, and links back to writing and working with the equations from earlier lessons.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing

Connection	Cost Comparison: read the break-even point of two cost lines — its x (how many) and y (the cost).	~12 min
Movement	Slide to the crossing: drag a marker and watch the gap between two plans close to zero at the break-even.	~12 min
Reflection	What does the crossing mean?: which option is cheaper for a few units, and which for many.	~10 min
Creativity	Which deal wins?: find a real break-even, then advise for light use and heavy use.	~6 min
Rest	A pause after seeing a single point settle a real decision.	~3 min
Nutrition	Contextual (Mode A): buy vs subscribe, membership vs pay-per-use — the break-even tells you which side you are on.	~2 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Reading only the x (or only the y) of the crossing

Repair: “The break-even has two parts — how many units (x) AND the cost there (y). What are both?”

Thinking one option is always cheaper

Repair: “Which is cheaper for a few units? And for many? The winner swaps at the crossing.”

Confusing 'cheaper to start' with 'cheaper overall'

Repair: “A low starting cost wins early, but a high rate can overtake it. Where do they meet?”

Thinking lines with different gradients might never meet

Repair: “Only parallel (equal-gradient) lines never meet. Different gradients always cross exactly once.”

Assuming the crossing is where a line meets an axis

Repair: “The break-even is where the two lines meet EACH OTHER, not where one meets an axis.”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line

Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: working out a cost from an equation (substituting a value of x), reading coordinates, and knowing that at a crossing the two lines share the same point. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Keep both coordinates and both sides in view:

- “Where do the two lines cross? How many units is that (x)?”
- “What is the cost at the crossing (y)? Check it in both plans.”
- “For a few units, which option is cheaper? And for many?”
- “So who should choose which deal — someone who uses a little, or a lot?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

“Do we have to solve equations, or just read the graph?”

Reading the break-even off the graph is enough here. If a learner is ready, setting the two costs equal and solving (for example $x + 30 = 4x$) gives the same answer and is a good next step.

“Why does the cheaper option swap at the crossing?”

Before the break-even the option with the smaller starting cost is ahead; its higher rate then eats into that lead until, at the crossing, they are equal — and beyond it the lower-rate option pulls ahead. The crossing is exactly the tipping point.

“Where does this show up in real life?”

Any 'pay once versus pay per use' decision: buy a printer or subscribe, gym membership or pay-per-visit, season ticket or single tickets. Businesses use the same idea to find where income overtakes costs — the point a product starts to make a profit.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.

Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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